

TILLAMOOK, OR, USA

WMO: 720202

Lat: 45.417N

Lon: 123.817W

Elev: 37

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 00118

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	27.6	29.9	14.0	11.2	36.1	21.1	15.8	39.4	28.4	54.8	24.9	49.7	3.4	140	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.8	80.8	62.4	74.6	62.1	71.8	61.5	65.6	74.2	64.0	70.5	62.7	68.4	7.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	87.4	65.7	61.4	81.4	65.2	60.7	79.3	64.7	30.3	74.9	29.2	71.3	28.4	68.0	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.4	18.5	16.1	DB	22.2	93.2	3.8	4.0	19.4	96.1	17.2	98.5	15.1	100.7	12.3	103.6
			WB	21.5	70.1	3.8	2.5	18.8	71.9	16.5	73.3	14.4	74.7	11.6	76.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	44.6	44.4	46.4	48.9	53.4	57.3	60.6	61.7	59.2	53.4	48.2	43.0
	DBStd	7.93	5.16	5.32	4.54	4.18	4.30	3.85	3.07	3.24	4.21	4.69	5.57	6.05
	HDD50	902	178	165	132	67	12	0	0	0	0	20	99	229
	HDD65	4845	632	576	578	484	361	234	140	112	181	360	505	682
	CDD50	1558	11	9	19	33	118	219	330	362	276	125	44	12
	CDD65	26	0	0	0	0	2	3	5	9	7	0	0	0
	CDH74	561	0	0	0	15	71	47	77	143	188	20	0	0
	CDH80	183	0	0	0	2	23	17	31	57	52	1	0	0
(14) Wind	WSAvg	4.5	5.1	4.8	5.2	5.0	4.7	4.8	4.9	3.8	3.3	3.4	4.5	5.1
(15) Precipitation	PrecAvg	91.8	14.4	10.5	9.7	6.7	4.6	3.3	1.4	1.7	3.5	7.4	13.5	14.9
	PrecMax	122.7	27.3	24.7	18.5	14.5	8.5	7.6	5.4	6.5	10.9	14.2	25.4	24.2
	PrecMin	61.6	0.9	2.3	1.2	2.2	0.6	0.7	0.0	0.0	0.1	0.6	2.6	4.2
	PrecStd	15.3	6.2	4.8	3.8	3.0	2.0	1.8	1.3	1.6	2.5	3.9	5.3	5.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.5	63.1	68.2	73.5	82.2	79.2	82.4	88.4	86.0	77.0	66.2	61.2
		MCWB	51.0	52.0	52.7	57.9	60.7	65.6	65.3	64.9	61.3	58.1	57.4	55.8
	2%	DB	57.0	58.8	61.2	65.9	72.0	71.7	73.0	76.8	80.6	69.8	62.6	57.2
		MCWB	51.1	50.0	51.9	55.1	58.8	62.2	62.8	64.2	62.2	56.9	56.1	52.8
	5%	DB	54.9	55.0	57.1	60.9	64.1	67.9	70.5	72.4	72.8	64.4	59.2	54.8
		MCWB	51.3	50.1	50.2	52.4	56.8	59.9	62.0	62.7	61.7	57.7	55.4	51.8
	10%	DB	53.8	53.6	54.5	56.8	61.3	64.5	68.4	70.0	68.4	62.8	57.1	53.6
		MCWB	50.9	50.1	50.0	51.0	54.8	57.9	60.8	61.8	59.9	57.3	53.9	51.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	55.3	56.0	59.4	63.4	66.8	66.4	67.0	67.9	64.3	60.6	57.9
		MCDB	57.2	60.6	62.0	71.2	76.1	77.6	76.4	77.5	75.4	67.0	62.4	60.4
	2%	WB	54.1	53.3	53.7	56.0	59.5	63.0	64.3	65.2	64.9	61.7	58.2	54.7
		MCDB	55.4	55.4	57.6	63.5	69.2	69.1	71.4	73.8	71.8	64.2	60.7	56.6
	5%	WB	52.5	51.8	52.4	53.8	57.2	60.8	62.9	63.7	63.1	59.9	56.5	52.9
		MCDB	53.9	53.9	55.8	57.9	64.1	66.0	68.6	69.9	68.9	62.9	59.1	54.4
	10%	WB	50.5	50.2	50.9	52.3	55.6	59.2	61.7	62.4	61.4	58.3	54.6	50.5
		MCDB	52.4	52.3	54.1	55.9	61.5	64.0	67.5	68.3	66.9	61.7	56.9	52.4
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	13.6	15.4	15.5	15.4	15.5	17.0	17.8	21.0	18.2	14.7	13.1
		MCDBR	17.3	19.0	20.9	23.7	23.2	21.3	21.3	24.8	30.5	23.4	16.8	13.6
		MCWBR	11.5	12.2	13.1	14.6	13.9	12.5	12.2	13.5	16.0	14.1	11.4	10.1
	5% WB	MCDBR	12.0	13.9	17.0	19.1	20.3	18.2	18.3	20.3	22.4	15.6	13.2	10.9
		MCWBR	9.6	10.5	11.8	12.7	12.5	10.9	10.7	11.5	13.3	11.0	10.1	9.6
(40) Clear-Sky Solar Irradiance	taub		0.326	0.322	0.337	0.366	0.378	0.364	0.349	0.357	0.348	0.336	0.330	0.328
	taud		2.509	2.545	2.511	2.393	2.386	2.441	2.508	2.489	2.517	2.546	2.514	2.472
	Ebn at Noon		249	275	285	284	283	286	290	283	277	265	245	233
	Edh at Noon		22	25	29	36	37	35	32	32	29	24	21	20
(44) All-Sky Solar Radiation	RadAvg		381	650	961	1384	1727	1891	2069	1771	1370	813	444	314
	RadStd		56	83	115	119	156	157	155	128	114	94	57	45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+40

Nomenclature: See separate page